Harsha Nimmagadda

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SUMMARY

Motivated engineer with a Master of Science in Electrical Engineering, specializing in . Project experience includes data analysis, design optimization, data validation, physical design on 7nm, 12nm, 15nm and 16nm. Eager to learn new tool technologies and seeking full-time opportunity to handle tough challenges in the semiconductor industry starting from May 2026.

EDUCATION

Doctor of Philosophy (PhD) in Computer Engineering Present
Arizona State University, Tempe, AZ GPA: 3.57
Course work: VLSI design, Advanced computer architecture, CMOS digital systems & circuits, Semiconductor device physics.

Master of Science in Computer Engineering May 2023
Arizona State University, Tempe, AZ GPA: 3.57

SKILLS

Technology node and programming skills: 7nm, 12nm, 15nm, 16nm, 32nm, Python, Verilog, Labview, TCL, Matlab.
Development skills: Block level synthesis, DFT, STA, CMOS, CDC, lint, Floorplan, PnR, CTS, Ultrafast design methodology, low-power design techniques, Power analysis, Signoff analysis, BIST, Qor assessment, Design automation, Physical verification.
EDA tools and OS: Cadence (Genus, Innovus, Virtuoso), Synopsys (Design Compiler NXT, ICC2, Primetime, VC Spyglass), Mentor Graphics (ModelSim, Calibre), Conformal, IC Validator, Xilinx Vivado, Windows, Linux, GTK wave, Microsoft office.
Certification: Synopsys Purple Certification - Physical Design.

EXPERIENCE

Systems Engineer | Cyber Nirvana Sep 2024 – Jun 2025
- Developed smart power distribution system and integrated PLC that enhances efficiency, reliability and load balancing.
- Deployed a load-shedding automation system to optimize power usage during peak hours, improving efficiency by 25%.
- Ensured power system simulations using MATLAB/Simulink to validate system performance before deployment.

IC Design Engineer | Analog Rails Feb 2024 – Aug 2024
- Designed standard cells, memory bitcells at transistor level, gate level in TSMC 16nm process using in-house EDA tools.
- Integrated Design-for-Test (DFT) logic such as scan chains to enable effective testability of 6T and 8T SRAM designs.
- Worked on memory complier of dual port SRAM 8T and developed testbench to test the functionality in SPICE mode.

IC Design Intern | Analog Rails Dec 2023 – Jan 2024
- Designed standard cells, high speed flops at transistor level, gate level in TSMC 16nm process using in-house EDA tools.
- Performed APL generation for standard cell libraries and extracted PPA data with better efficiency on 12nm node.
- Implemented Std cell layout physical verification (DRC), fixed bugs to meet PDK runset and block level requirements.

Graduate Research Assistant | Arizona State University May 2022 – May 2023
- Developed secure DFT strategies for and algorithm development of hardware security ASIC design to prevent leakage.
- Verified scan chain connectivity and functionality using RTL simulation and post-synthesis netlist checks on 15 nm node.
- Strengthened incremental floorplan at chip level, block level and implemented PnR to achieve max frequency 250 Mhz.

Data Analyst | Cognizant Technology Solutions Jun 2020 - Jul 2021
- Implemented GSD management using python to retrieve data and processed data to analyze behavior of design.
- Collaborated with cross- functional team to identify root cause for customer queries and achieved 95% efficiency.

ACADEMIC PROJECTS

ASIC Acceleration for Graph Convolution Networks (GCN) using 7nm ASAP PDK Spring 2023
- Designed a hardware accelerator for Graph Convolution Network in verilog to perform machine lvearning on graph data.
- Implemented RTL to GDS flow on the design and verified DRC and LVS checks by importing GDS into virtuoso.
- Optimized the design using pipelining techniques and finally achieved area (300 µm^2), power (390 µW) of the design.

RTL to GDSII Implementation: 2-Bit Adder using 7nm ASAP PDK Spring 2023
- Collaborated with team and implemented 2-bit adder block using verilog and verified the functionality in modelsim.
- Performed RTL logic synthesis using synopsys design compiler to generate gate level netlist, power and area reports.
- Experimented floorplan and implemented automatic place and route - APR using cadence innovus to generate GDSII.

Custom Standard Cell Library Design in 7 nm and 32 nm PDK Fall 2021
- Developed custom layout for NAND and inverter circuit design from gate level to a layout in 32nm and 7nm.
- Assessed design simulation to perform DRC, LVS checks by debugging and involved parasitic extraction of standard cell.